Robust Real Rate Rules

Avoiding self-fulfilling fluctuations

based on Holden (2024)

Warsaw Reading Group, July 2026

Table of contents

1. Implementing an inflation target
2. Robustness results
3. Practice and evidence

Motivation

When monetary policy rules are discussed, we choose Taylor rule.
Problems:

- hand-to-mouth households;
- high government spending;
- non-rational expectations;

and many more!

Question: How to make the best monetary policy rule, that would be robust to these problems?

Solution

Instead of targeting interest rate based on inflation and output gap, we use real rate:

$$i_t = r_t + \phi\pi_t$$

or in the smoothed version:

$$i_t - r_t = (i_{t-1} - r_{t-1}) + (\mathbb{E}_t\pi^*_{t+1} - \mathbb{E}_{t-1}\pi^*_t) + \theta(\pi_t - \pi^*_t)$$

and that's it!

Working through Fisher equation

That is because the monetary rule:

$$i_t = r_t + \phi\pi_t$$

and the Fisher equation:

$$r_t = i_t - \mathbb{E}_t\pi_{t+1}$$

yield one result:

$$\phi\pi_t = \mathbb{E}_t\pi_{t+1}$$

which for any $\phi > 1$ ensures determinacy of inflation at $\pi_t = 0$, regardless of any other parameters and equations. The same is true for the target inflation - the result is that $\pi_t = \pi_t^*$ is the equilibrium.

How monetary policy is conducted?

The monetary policy is possible thanks to the changing of the inflation target:

$$i_t = r_t + \mathbb{E}_t \pi^*_{t+1} + \phi(\pi_t - \pi^*_t)$$

Suppose there is a model with NKPC:

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + \kappa \omega_t$$

Central bank minimizes $\mathbb{E}_0 \sum_t \beta^t [\pi_t^2 + \lambda x_t^2]$.
This requires $\pi_t = -\kappa^{-1}\lambda(x_t - x_{t-1})$ (to minimise social losses).

Setting $\pi^*_t := -\kappa^{-1}\lambda(x_t - x_{t-1})$ and plugging into the real rate rule implements optimal policy.

Monetary shocks

Add a monetary policy shock ζ_t (e.g. from imperfect information):

$$i_t = r_t + \phi\pi_t + \zeta_t$$

Combined with the Fisher equation:

$$\mathbb{E}_t\pi_{t+1} = \phi\pi_t + \zeta_t$$

If ζ_t is AR(1) with persistence ρ, the unique stationary solution is

$$\pi_t = -\frac{1}{\phi - \rho}\zeta_t$$

So central bank can, if it chooses, reduce inflation without changing an equilibrium solution (AR(1) process dies out).

Fisher equation wedges

Suppose the Fisher equation holds only up to a (possibly endogenous) wedge ν_t:

$$i_t = r_t + \mathbb{E}_t \pi_{t+1} + \nu_t$$

Source: risk premia, liquidity/convenience yield, bounded rationality

Under mild boundedness conditions on ν_t, for $\phi > 1$:

$$|\mathbb{E}\pi_t| = O\left(\frac{1}{\phi}\right), \qquad \mathrm{Var}(\pi_t) = O\left(\frac{1}{\phi^2}\right) \quad \text{as } \phi \to \infty$$

Alternative: target inflation swap rates directly, which may be less distorted than bond prices.

Zero Lower Bound

Naive real rate rule breaks at the ZLB: $i_t - r_t$ can no longer be freely set once $i_t = 0$ binds.

Fix: modify the target so nominal rates stay positive:

$$\check{\pi}_t^* := \max\{\pi_t^*,\ \varepsilon - r_{t-1}\}, \qquad \varepsilon > 0 \text{ small}$$

and use the smoothed rule with $\check{\pi}_t^*$ in place of π_t^*:

$$i_t = \max\left\{0,\ r_t + (i_{t-1} - r_{t-1}) + \left(\mathbb{E}_t \check{\pi}_{t+1}^* - \mathbb{E}_{t-1} \check{\pi}_t^*\right) + \theta(\pi_t - \check{\pi}_t^*)\right\}$$

And this delivers a unique closed-form solution $\pi_t = \check{\pi}_t^*$, rules out deflationary bias and severe ZLB deflations and with smoothing, only $\theta > 0$ (not > 1) is needed to **rule out sunspot-driven jumps** into the ZLB.

Practical application

Real-world frictions: inflation-protected bonds are long maturity (T periods), inflation is released with a lag (S), and real bonds have an indexation lag (L).

Generalized Fisher equation:

$$i_{t|t-S} = r_{t|t-S} + \nu_{t|t-S} + \mathbb{E}_{t-S}\frac{1}{T}\sum_{k=1}^{T}\pi_{t+k-L}$$

- extend the smoothed, modified-target rule to this T-period setting;
- determinacy still holds for any $\theta > 0$;
- as θ grows large, $\pi_t \approx \check{\pi}_t^*$ even with unobserved, non-stationary wedges;
- using longer bonds also makes hitting the ZLB less likely.

Practical takeaway: central banks can implement this with existing liquid markets (TIPS-Treasury spreads, or inflation swaps).

Empirical testing

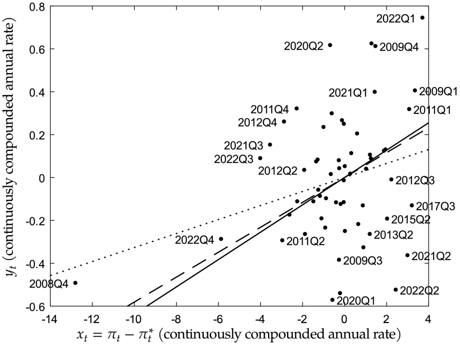

Figure: y_t is the change in five-year breakeven inflation expectations, relative to target, minus 5% of the change in inflation, relative to target. The dotted line is the OLS estimate. The dashed line is the monetary shock-based estimate. The solid line is the IV estimate.

Conclusions

Under a robust real rate rule:

- the central bank can always hit its inflation target, regardless of the rest of the economy;
- inflation moves only if the central bank is insufficiently aggressive, or chooses to move it;
- causation runs from inflation to the output gap, never the reverse;
- household/firm heterogeneity, rationality, and fiscal policy are (largely) irrelevant for inflation dynamics;
- with a time-varying short-term target, the rule can implement optimal policy;
- robust to the ZLB, bounded rationality, Fisher equation wedges, and active fiscal policy.